THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO SECONDARY SCHOOL

FORM FOUR PRE-MOCK EXAMINATION
041 BASIC MATHEMATICS
MARKING SCHEME APRIL 2025

SECTION A (60 Marks)

1. (a) i. 0.034500 has five (5) significant figures.....01 mark

ii. 69 500 kg has five (5) significant figures.....01 mark

iii. $40^2 = 40 \times 40 = 1\,600$.....$00\frac{1}{2}$ mark

Therefore 40^2 has two (2) significant figures.....$00\frac{1}{2}$ mark

(b) $\sqrt{256} = 16, 2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32, 2 \times 2 \times 2 \times 3 = 24$.....$01\frac{1}{2}$ m arks

2	16	32	24
2	8	16	12
2	4	8	6
2	2	4	3
2	1`	2	3
3	1	1	3
	1	1	1.....$00\frac{1}{2}$ mark

GCF $= 2^3 = 8$, LCM $= 2^5 \times 3 = 96$ .....01 mark

2. (a) $\left(\frac{\sqrt{2}-\sqrt{3}}{\sqrt{3}+\sqrt{2}}\right)^2 = \left(\frac{\sqrt{2}-\sqrt{3}}{\sqrt{3}+\sqrt{2}}\right)\left(\frac{\sqrt{2}-\sqrt{3}}{\sqrt{3}+\sqrt{2}}\right)$

$= \frac{2-\sqrt{6}-\sqrt{6}+3}{3+\sqrt{6}+\sqrt{6}+2}$

$= \frac{5-2\sqrt{6}}{5+2\sqrt{6}}$ .....01 mark

$= \frac{5-2\sqrt{6}}{5+2\sqrt{6}} \times \frac{5-2\sqrt{6}}{5-2\sqrt{6}}$.....$00\frac{1}{2}$ mark

$= \frac{25-10\sqrt{6}-10\sqrt{6}+24}{25-10\sqrt{6}+10\sqrt{6}-24}$.....$00\frac{1}{2}$ mark

$= \frac{25+24-20\sqrt{6}}{25-24}$

$= 49 - 20\sqrt{6}$.....01 mark

(b) $\frac{\log_2 2^7 - \log_2 2^3}{\log_2 x} = \log_2 x$

$\frac{7\log_2 2 - 3\log_2 2}{\log_2 x} = \log_2 x$ .....$00\frac{1}{2}$ mark

$\frac{4}{\log_2 x} = \log_2 x$

$4 = (\log_2 x)^2$.....01 mark

$\sqrt{4} = \sqrt{(\log_2 x)^2}$

$\pm 2 = \log_2 x$.....$00\frac{1}{2}$ mark

$2^2 = x$

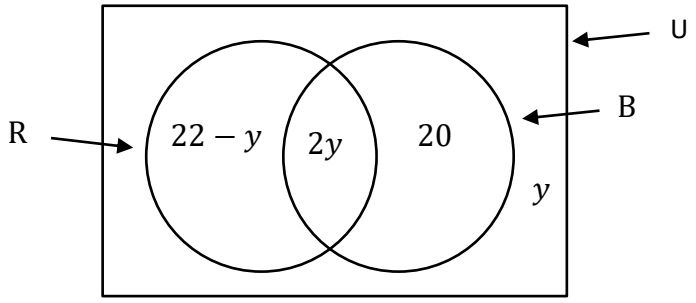
$x = 4$…….. ……01 mark

3. (a) Let R be the number of guests who ate rice and B the number of guests who ate bananas

Given: $n(\mu) = 45$, $n(R) = 22$, $n(R \cap B) = 2y$, $n(R' \cap B')or\ n(R' \cup B') = y$, and , $n(R' \cap B) = 20$

By using a Venn diagram

Figure 1



$22 + 2y - 2y + y + 20 = 45$……$00\frac{1}{2}$ mark

$y = 45 - 42$

$y = 3$……01 mark

Guests who ate bananas only $= 22 - 2y$

$= 22 - 2 \times 3$

$= 22 - 6$

$= 16$

Therefore 16 guests ate bananas only……01 mark

(b) Let A be the number of cards multiple 3 and B be number of cards multiple 5

A $= \{3 , 6 , 9 , 12 , 15 , 18 , 21 , 24 , 17 , 30\}$……$00\frac{1}{2}$ mark ….

$B = \{5 , 10 , 15 , 20 , 25 , 30\}$……$00\frac{1}{2}$ mark ……

$A \cap B = \{15 , 30\}$……$00\frac{1}{2}$ mark

$n(A) = 10, n(B) = 6 , n\ (A \cap B) = 2$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= \frac{10}{30} + \frac{6}{30} - \frac{2}{30}$……$00\frac{1}{2}$ mark.

$= \frac{16}{30} - \frac{2}{30}$

$= \frac{14}{30} = \frac{14 \div 2}{30 \div 2} = \frac{7}{15}$……01 mark

4. (a) From distance formula

$d = \sqrt{(x_2 - x_1)^2 + \ (y_2 - y_1)^2}$

$13 = \sqrt{(4 + 1)^2 + (y - 7)^2}$

$13 = \sqrt{5^2 + y^2 - 14y + 49}$

$13^2 = \left(\sqrt{y^2 - 14y + 74}\right)^2$

$169 = y^2 - 14y + 74$……$00\frac{1}{2}$ mark

$y^2 - 14y = 74 - 169 = 0$

$y^2 - 14y - 95 = 0$ ……equation 1…… ……01 mark

Solve equation 1 by general formula

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{14 \pm \sqrt{14^2 - 4 \times 1 \times -95}}{2}$$

$$= \frac{14 \pm \sqrt{196 + 380}}{2}$$ ……$00\frac{1}{2}$ mark

$$= \frac{14 \pm \sqrt{576}}{2}$$

$$= \frac{14 \pm 24}{2}$$

$$= \frac{14+24}{2} \; or \; \frac{14-24}{2}$$

$$= \frac{38}{2} \; or \; \frac{-10}{2}$$

Therefore $y = 19 \; or \; y - 5$……01 mark

(b) i. $\overrightarrow{QP} = -a$(the negative of vector $\overrightarrow{PQ}$) …… ……$00\frac{1}{2}$ mark

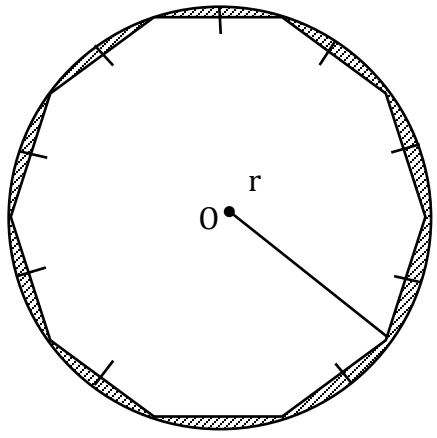
ii. $\overrightarrow{SP} = -b$(The negative of vector $\overrightarrow{QR}$……$00\frac{1}{2}$ mark.

iii. $\overrightarrow{RP} = \overrightarrow{PQ} + \overrightarrow{QR} = 0$ (Resultant of vectors$\overrightarrow{PQ} + \overrightarrow{QR}$)…… …… ……01 mark

iv. $\overrightarrow{PR} = \overrightarrow{PQ} + \overrightarrow{QR} = \underline{a} + \underline{b}$ (Resultant of vectors$\overrightarrow{PQ} \; and \; \overrightarrow{QR}$) or triangular law……01 mark

5. (a) Sketching of the regular decagon

Figure 2



Given that $n = 10, s = 3.09 \; cm$

The length or the side of regular decagon inscribed in a circle is given by

$$s = 2rsin\left(\frac{180^\circ}{n}\right)$$

$$3.09 \; cm = 2rsin\left(\frac{180^\circ}{10}\right)$$

$$3.09 = 2rsin \; 18^\circ$$

$r = \frac{3.09\ cm}{2\sin 18^{\circ}} = 4.9997\ \ldots$

$r \approx 5\ cm$.....01 mark

Area of a circle $(A_1) = \pi r^2 = 3.14\ \times 5^2 = 78.5 cm^2$.....$00\frac{1}{2}$ mark

Area of a regular decagon $(A_2) = \frac{1}{2} n r^2 \sin\left(\frac{360^{\circ}}{n}\right)$

$= \frac{1}{2} \times 10 \times (5\ cm)^2 \sin\left(\frac{360^{\circ}}{10}\right)$

$= 125 cm^2 \sin 36^{\circ}$

$= 73.47\ cm^2$.....$00\frac{1}{2}$ mark

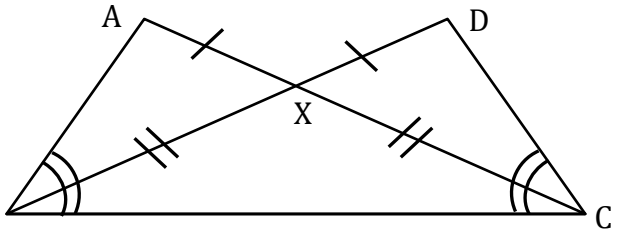
Area of the shaded region $= A_1 - A_2$

$= 78.5\ cm^2 - 73.47\ cm^2$

$= 5.03\ cm^2$.....01 mark

(b)Solution

Figure 3



$\overline{AX} = \overline{DX}$ (Given)..... $00\frac{1}{2}$ mark

$\overline{BX} = \overline{CX}$(Given) .....$00\frac{1}{2}$ mark

$\overline{BC} = \overline{BC}$ (Is common)..... $00\frac{1}{2}$ mark

$\angle ABC = \angle DCB$ (Given)..... $00\frac{1}{2}$ mark

Therefore $\angle BAC = \angle CDB$(ASA axioms).....01 mark

6. (a) Total distance walked by Seypalm's friend $= (5.2\ km + 1.8\ km) \times 2 \times 5$.....01 mark

$= 7\ km\ \times 10$.....01 mark

$= 70\ km$.....01 mark

Therefore, he or she walks 70 km in 5 days

(b) $z \propto \frac{\sqrt{x}}{y}$, $z = \frac{\sqrt{x}}{y} k, k = \frac{zy}{\sqrt{x}}\ or\ k = \frac{zy\sqrt{x}}{x}$ .....01 mark

$x_1 = 3, y_1 = 4, z_1 = 4.5, x_2 = 6, y_2 = 9, required\ z_2$

$k = \frac{4.5\times 4}{\sqrt{3}} = \frac{18}{\sqrt{3}} = \frac{18}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{18\sqrt{3}}{3} = 6\sqrt{3}$ .....01 mark

$6\sqrt{3} = \frac{z\times 9}{\sqrt{6}}$

$z = \frac{\sqrt{6}\times 6\sqrt{3}}{9} = \frac{6\sqrt{18}}{9} = \frac{6\times 3\sqrt{2}}{9} z = 2\sqrt{2}$.....01 mark

7. (a) Total parts $= 9 + 5 + 4 + 2 = 20$ parts.....01 mark

Grace's share $= \frac{9}{20} \times 80\ 000$ shillings

$= 36\ 000$ shillings.....01 mark

30% of 36 000 shillings $= \frac{30}{100} \times 36\,000 = 10\,800$ shillings ……01 mark

(b) Solution

Income statement for the year ended 31st December 2019

Sales		2 309 000
Less: cost of goods sold		
Purchase		2 267 000
Gross profit		42 000
Less : expenses		
Rent	114 000	
Salaries	67 000	
Wages	54 000	
Total expenses		235 000
Net loss		193 000

Statement of financial position as at 31st December 2019

Non –Current Assets (Fixed Assets)		
Furniture		305 000
Shelves		270 000
		575 000
Current Assets		
Cash		1 097 000
		2 482 000
Financed by		
Capital		2 675 000
Less net loss		193 000
		2 482 000

……..03 marks

8. (a) $A_3 = A_1 + 2d = 12$

$A_1 + 2d = 12$ ……equation 1

$A_n = A_{50} = A_1 + 49d = 106$ ……equation 2

Solve equation 1 and 2 by elimination method

$$-\begin{vmatrix} A_1 + 2d = 12 \\ A_1 + 49d = 106 \end{vmatrix}$$

$47d = 94$, $\frac{47d}{47} = \frac{94}{47}$, $d = 2$ .....01 mark

Substitute 2 for d in $A_1 + 2d = 12$

$A_1 + 2 \times 2 = 12$

$A_1 = 12 - 4 = 8$ .....01 mark

$S_{50} = \frac{n}{2}(A_1 + A_N)$

$= \frac{50}{2}(8 + 106)$

$= 25 \times 114$

$= 2\,850$ .....01 mark

(b) $r = \frac{G_2}{G_1} = \frac{G_3}{G_2} = \frac{6}{2} = \frac{18}{6} = 3$ .....01 mark

$G_n = G_1 r^{n-1}$

$1000 = 2 \times 3^{n-1}$

$\frac{10\,000}{2} = \frac{2}{2} \times 3^{n-1}$

$5000 = 3^{n-1}$

Introduce logarithm both sides

$\log 5000 = \log 3^{n-1}$

$n = \frac{\log 5000}{\log 3} + 1$.....$00\frac{1}{2}$ mark

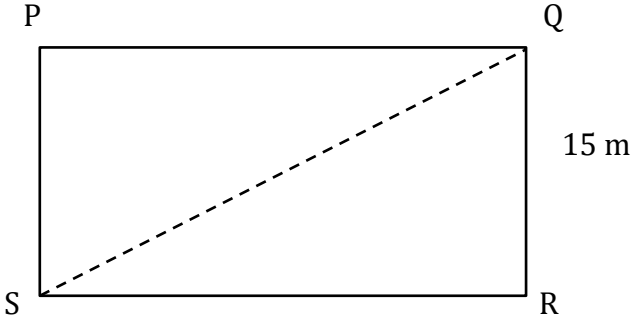
$n = 8.752 \ldots$ .....$00\frac{1}{2}$ mark

$n \approx 9$.....$00\frac{1}{2}$ mark

So , the first term to exceed 10 000 is $G_9 = 13122$.....$00\frac{1}{2}$ mark.

9. (a)

Figure 4



$A = l \times w$, $300\ m^2 = 15m \times l$

$l = \frac{300\ m^2}{15\ m} = 20\ m$ .....01 mark

$\overline{SQ} = \sqrt{(20\ m)^2 + (15\ m)^2}$

$= \sqrt{400\,m^2 + 225\,m^2}$

$= \pm\sqrt{625\,m^2}$

$= \pm 25$

Therefore diagonal $= \overline{SQ} = 25\,m, \overline{SQ} \geq 0$ .....01 mark

(b)$(1 - \cos 30^\circ)(1 - \cos 210^\circ) = (1 - \cos 30^\circ)(1 - \cos(180^\circ + 30^\circ))$

$(1 - \cos 30^\circ)\left(1 - (\cos 180^\circ \cos 30^\circ - \sin 180^\circ \sin 30^\circ)\right)$

$= (1 - \cos 30^\circ)(1 - (-1 \cos 30^\circ - 0 \sin 30^\circ)$.....$00\frac{1}{2}$ mark.

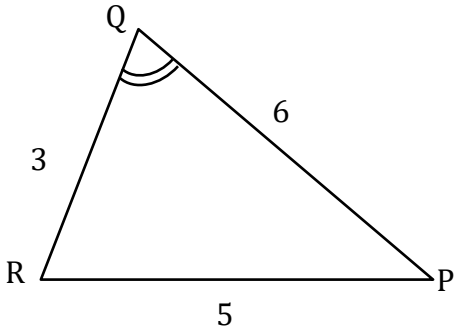
$= (1 - \cos 30^\circ)(1 + \cos 30^\circ)$

$= 1 - cos^2 30^\circ = sin^2 30^\circ$.....$00\frac{1}{2}$ mark.

$= 1 - \left(\frac{\sqrt{3}}{2}\right)^2$.....$00\frac{1}{2}$ mark.

$= 1 - \frac{3}{4} = \frac{1}{4}$.....$00\frac{1}{2}$ mark

(c) Figure 5



$00\frac{1}{2}$ mark

$5^2 = 3^2 + 6^2 - 2 \times 3 \times 6 \cos\theta$.....$00\frac{1}{2}$ mark

$25 = 9 + 36 - 36 \cos\theta$

$\text{Cos}\,\theta = \frac{45-25}{36}$.....$00\frac{1}{2}$ mark

$= \frac{20}{36} = \frac{20 \div 4}{36 \div 4} = \frac{5}{9}$.....$00\frac{1}{2}$ mark

10. (a) $3x^2 + 8x + 5 = 0$

$\frac{3x^2}{3} + \frac{8x}{3} + \frac{5}{3} = \frac{0}{3}$

$x^2 + \frac{8x}{3} + \frac{5}{3} = 0$

$x^2 + \frac{8}{3}x + \frac{5}{3} - \frac{5}{3} = 0 - \frac{5}{3}$

$x^2 + \frac{8}{3}x = -\frac{5}{3}$.....01 mark

$.x^2 + \frac{8}{3}x + \frac{64}{36} = -\frac{5}{3} + \frac{64}{36}$

$$\left(x+\frac{8}{6}\right)^2=\frac{-60+64}{36}$$

$$\left(x+\frac{8}{6}\right)^2=\frac{4}{36}$$

$$\sqrt{\left(x+\frac{8}{6}\right)^2}=\pm\sqrt{\frac{4}{36}}$$

$x+\frac{8}{6}=\pm\frac{2}{6}$.....01 mark

$x=\frac{-8}{6}\pm\frac{2}{6}$

$x=\frac{-8}{6}+\frac{2}{6} or\ x=\frac{-8}{6}-\frac{2}{6}$

$x=\frac{-6}{6} or\ x=\frac{-10}{6}$

$x=-1\ or\ x=\frac{-5}{3}$.....01 mark

(b) Area of a square ($A_1=s\times s=s^2$

Area of the increased square

$A_2=(s+4\ cm)(s+4\ cm)=s^2+40\ cm$ .....01 mark

$S^2+8s+16=s^2+40\ cm\ 16$

$8\ s=40\ cm-16\ cm$

$8S=24\ cm$ .....01 mark

$s=\frac{24\ cm}{8\ cm}=3\ cm$

Therefore the side of the original square is 3 cm.....01 mark

SECTION B (40 Marks)

11. (a)

Class intervals	cf	f	x	fx
$125-130$	5	5	127.5	637.5
$131-136$	13	8	133.5	1 068
$137-142$	25	12	139.5	1 674
$143-148$	29	4	145.5	582
$149-154$	30	1	151.5	151.5
		30....$00\frac{1}{2}$ mark.	.....$00\frac{1}{2}$ mark..	$\sum fx$ $=4\ 113.$ ....01 mark

i. Mean ($\overline{x}$) $=\frac{\sum fx}{\sum f}=\frac{4\ 113}{30}$.....$00\frac{1}{2}$ mark

$=137.1$.....01 mark

ii. Mode $= l + \left(\frac{t_1}{t_1+t_2}\right) i$, where by modal class is $137 - 142$

$Mode = 136.5 + \left(\frac{12-8}{((12-8)+(12-4))}\right) 6$

$= 136.5 + \left(\frac{4}{12}\right) 6$.....$00\frac{1}{2}$ mark.

$= 136.5 + 2$

$= 138.5$.....01 mark

iii. Median class is $137 - 142$

$median = l + \left(\frac{\frac{N}{2} - \sum fb}{fw}\right) i$

$= 136.5 + \left(\frac{\frac{30}{2} - 13}{12}\right) 6$

$= 136.5 + \left(\frac{15-13}{12}\right) 6$

$= 136.5 + \frac{2}{12} \times 6$.....$00\frac{1}{2}$ mark

$= 136.5 + 1$

$= 137.5$.....01 mark

(b)

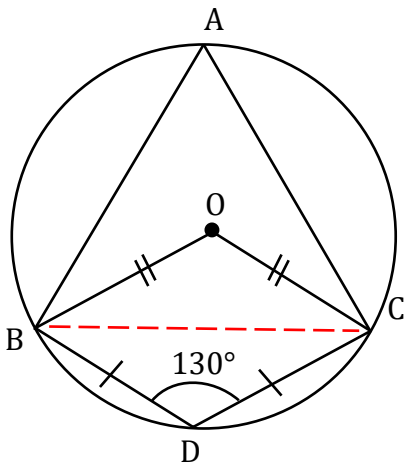


Figure 6

i. the polygon is a kite.....01 mark

ii . $\overline{OC} = \overline{OB}$ (Radii), construct segment $\overline{BC}$

$\angle BDC = \frac{1}{2}\angle BOC\ (reflex)$

$\angle BOC\ (reflex) = 2 \times 130^{\circ} = 260^{\circ}$.....$00\frac{1}{2}$ mark

$\angle BOC(reflex) + \angle BOC\ (obtuse) = 360^{\circ}$

$260^{\circ} + \angle BOC(\ obtuse) = 360^{\circ}$

$\angle BOC\ (\ obtuse) = 360^{\circ} - 260^{\circ} = 100^{\circ}$.....$00\frac{1}{2}$ mark.

From ΔBOC , $\angle OBC = \angle OCB$ (base angles of isosceles triangle)

$2\angle OCB + 100^{\circ} = 180^{\circ}$ (Sum of interior angles of a triangle)

$\angle OCB = \frac{180^{\circ} - 100^{\circ}}{2} = \frac{80^{\circ}}{2} = 40^{\circ}$ ……$00\frac{1}{2}$ mark

Again From ΔDBC , $\angle DBC = \angle DCB$ (base angles of isosceles triangle)

$2\angle DCB + 130^{\circ} = 180^{\circ}$ (Sum of interior angles of a triangle)

$\angle DCB = \frac{180^{\circ} - 130^{\circ}}{2} = \frac{50^{\circ}}{2} = 25^{\circ}$ ……$00\frac{1}{2}$ mark

$\angle OCD = \angle DCB + \angle OCB$

$= 40^{\circ} + 25^{\circ} = 65^{\circ}$ ……..01 mark

12. (a) Angular distance $= 45^{\circ} - 39^{\circ} = 6^{\circ}$……..01 mark

Distance in nautical miles $= 6 \times 60' = 360'$……01 mark

Speed $= \frac{distance}{time}$

$= \frac{360\ '}{3\ 0\ hours} = 12$ Nautical miles per hour or 12 knots……01 mark

Distance in kilometres $= 6\ \times 60' \times 1.852km = 666.72\ km$ ……..01 mark

Speed $= \frac{distance}{time} = \frac{666.72\ km}{30\ hour} = 22.224\ km\ per\ hour$ ……01mark

(b) i. $\overline{BD} = \overline{AC}$ (diagonal of a rectangle)

From ΔABC,

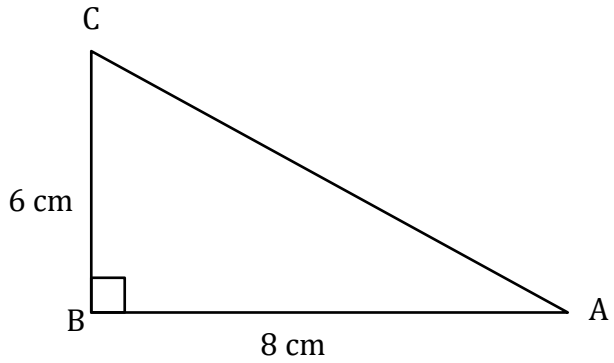


Figure 7

$\overline{CA} = \overline{BD} = \sqrt{6^2 + 8^2}$

$= \sqrt{36 + 64} = \pm\sqrt{100} = \pm 10\ cm$

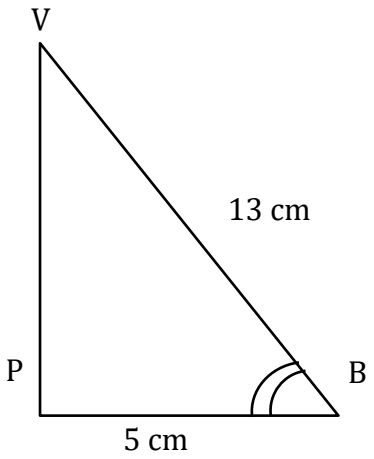
Therefore $\overline{BD} = \overline{AC} = 10\ cm, \overline{AC} \geq 0$……01 mark

From ΔVBP

$\overline{PB} = \frac{1}{2}\overline{BD} = 5\ cm$……$00\frac{1}{2}$ mark

$\overline{VP} = \sqrt{13^2 - 5^2}$

Figure 8



$$= \sqrt{169 - 25} = \pm\sqrt{144} = \pm 12\ cm$$

Therefore height $\overline{VP} = 12\ cm, \overline{VP} \geq 0$……01 mark

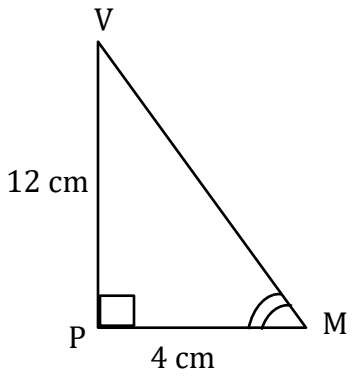
ii. From triangle VBP

$$\angle VBP = Tan^{-1}\left(\frac{12}{5}\right) = 67^{\circ}23'$$

Therefore the angle between $\overline{VB}$ and the base ABCD IS $67^{\circ}23'$……01 mark

iii. The angle between triangle VBC and the base ABCD is angle VMP. The line VP and MP are both perpendicular to line BC and M is the midpoint of line BC. Consider a right-angled triangle VMP in the pyramid

Figure 9



$$\overline{PM} = \frac{1}{2}\overline{AB} = 4\ cm\ \angle VMP = tan^{-1}\left(\frac{12}{4}\right) = 71^{\circ}34'$$

Therefore, the angle between triangle VBC and the base ABCD is $71^{\circ}34'$….01 mark

13. (a) $B\ C = A$

$C = AB^{-1}$……$00\frac{1}{2}$. mark

$|B| = 11 - 12 = -1$……$00\frac{1}{2}$. mark

$B^{-1} = \frac{1}{-1}\begin{pmatrix} 1 & -3 \\ -4 & 11 \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ 4 & -11 \end{pmatrix}$……$00\frac{1}{2}$. mark

$$C = \begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}\begin{pmatrix} -1 & 3 \\ 4 & -11 \end{pmatrix}$$

$$= \begin{pmatrix} -2+16 & 6-44 \\ -3+24 & 9-66 \end{pmatrix}$$

$$= \begin{pmatrix} 14 & -38 \\ 21 & -57 \end{pmatrix}$$ ……00$\frac{1}{2}$ mark

(b) let the number of boxes of marker pens be x and the number of boxes of correction fluid be y $\begin{cases} 50x + 20y = 40\,000 \\ 10x + 20y\ 24\,000 \end{cases}$ ……01 mark

Express the equation in matrix form

But $\begin{cases} 50x + 20y = 40\,000 \\ 10x + 20y = 24\,000 \end{cases}$ *is the same as* $\begin{cases} 5x + 2y = 4\,000 \\ x + 2y = 2\,400 \end{cases}$

$\begin{pmatrix} 5 & 2 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4\,000 \\ 2\,400 \end{pmatrix}$, let $\begin{pmatrix} 5 & 2 \\ 1 & 2 \end{pmatrix}$ be A

$|A| = 12 - 2 = 8$ ……00$\frac{1}{2}$ mark

$$A^{-1} = \frac{1}{8}\begin{pmatrix} 2 & -2 \\ -1 & 5 \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & \frac{-1}{4} \\ \frac{-1}{8} & \frac{5}{8} \end{pmatrix}$$ ……01 mark

$$\begin{pmatrix} 5 & 2 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} \frac{1}{4} & \frac{1-}{4} \\ \frac{-1}{8} & \frac{5}{8} \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{4} & \frac{-1}{4} \\ \frac{-1}{8} & \frac{5}{8} \end{pmatrix}\begin{pmatrix} 4\,000 \\ 2\,400 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{4} \times 4\,000 - \frac{1}{4} \times 2\,400 \\ -\frac{1}{8} \times 4\,000 + \frac{5}{8} \times 2\,400 \end{pmatrix}$$ ……00$\frac{1}{2}$ mark.

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1\,000 - 600 \\ -500 + 1\,500 \end{pmatrix}$$ ……00$\frac{1}{2}$ mark

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 400 \\ 1\,000 \end{pmatrix}$$

Therefore one box of marker pens costs 400 shillings while one box of correction fluid costs 1 000 shillings ……01 mark

(c) By using the formula $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$ *where* $k = 3$ *and* $(x, y) = (1, 2)$ *gives*

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$$ ……01 mark

Again $\begin{pmatrix} x'' \\ y'' \end{pmatrix} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix}\begin{pmatrix} x' \\ y' \end{pmatrix}$ *where* $k = 4$ *an*$(x', y')d$ $= (3,6)$ *gives*

$$\begin{pmatrix} x'' \\ y'' \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}\begin{pmatrix} 3 \\ 6 \end{pmatrix}$$ ……00$\frac{1}{2}$ mark

$= \begin{pmatrix} 12 \\ 24 \end{pmatrix}$ ……01 mark

14. (a) i. Equation of line of symmetry is given by the average of x −intercepts

$x = \frac{-b}{2a} = \frac{x_1+x_2}{2} = \frac{2+6}{2} = \frac{8}{2} = 4$……01 mark

ii. The parabola has equation in the form of$y = ax^2 + bx + c$

Intercepts are $(2\ ,\ 0), (6\ ,0), (0\ ,\ 5)$

From point $(\ 2\ ,\ 0)$ $0 = a(2^2) + 2b + 5$,since $c = y - intercept = 5$

$4a + 2b = -5$ ……equation 1

From point $(\ 6\ ,\ 0)$ $0 = a(6^2) + 6b + 5$,since $c = y - intercept = 5$

$36a + 6b = -5$ ……equation 2

Solve equation 1 and2 by elimination method

$$\begin{array}{l|l} 3 & 4a + 2b = -5 \\ 1 & 36a + 6b = -5 \end{array}$$

$$12a + 6b = -15$$
$$36a + 6b = -5$$

$$\begin{array}{l|l} - & 36a + 6b = -5 \\ & 12a + 6b = -15 \end{array}$$

$24a = 10, \frac{24a}{24} = \frac{10}{24},\ a = \frac{5}{12}$……$00\frac{1}{2}$ mark

Substitute $\frac{5}{12}$for $a\ in\ 4a + 2b = -5$

$4 \times \frac{5}{12} + 2b = -5$

$2b = -5 - \frac{5}{3} = \frac{-15-5}{3}$

$b = \frac{-20}{3} \times \frac{1}{2} = \frac{-10}{3}$……$00\frac{1}{2}$ mark

Therefore $f(x) = \frac{5}{12}x - \frac{10}{3} + 5$……$00\frac{1}{2}$ mark

Domain of $f = x: \{x \mathcal{E} \mathbb{R}\}$ ……$00\frac{1}{2}$ mark

Range of $f = y: \left\{y \geq f\left(\frac{-b}{2a}\right)\right\}$

$= \left\{y: y \geq \frac{5}{12}(4^2) - \frac{10}{3}(4) + 5\right\}$

$= \left\{y: y \geq \frac{5}{12} \times 16 - \frac{40}{3} + 5\right\}$

$= \left\{y: y \geq \frac{20}{2} - \frac{40}{3} + 5\right\}$……$00\frac{1}{2}$ mark

$= \left\{y: y \geq \frac{-20+15}{3}\right\}$

$= \{y: y \geq \frac{-5}{3}\}$

Therefore range $f = \{y: y \geq \frac{-5}{3}\}$……..……..$00\frac{1}{2}$ mark

(b) summarize the information into a table as follows:

	Grade A	Grade B	Total materials available
Starting materials	7 kg	5 kg	70 kg
Finishing materials	1 kg	3 kg	24 kg
Profit per grade of bar soaps	2 000	1 500	

Let the number of bar soap made from grade A *be* x
Let the number of bar soap made from grade B *be* y
Constraint or resources restrictions

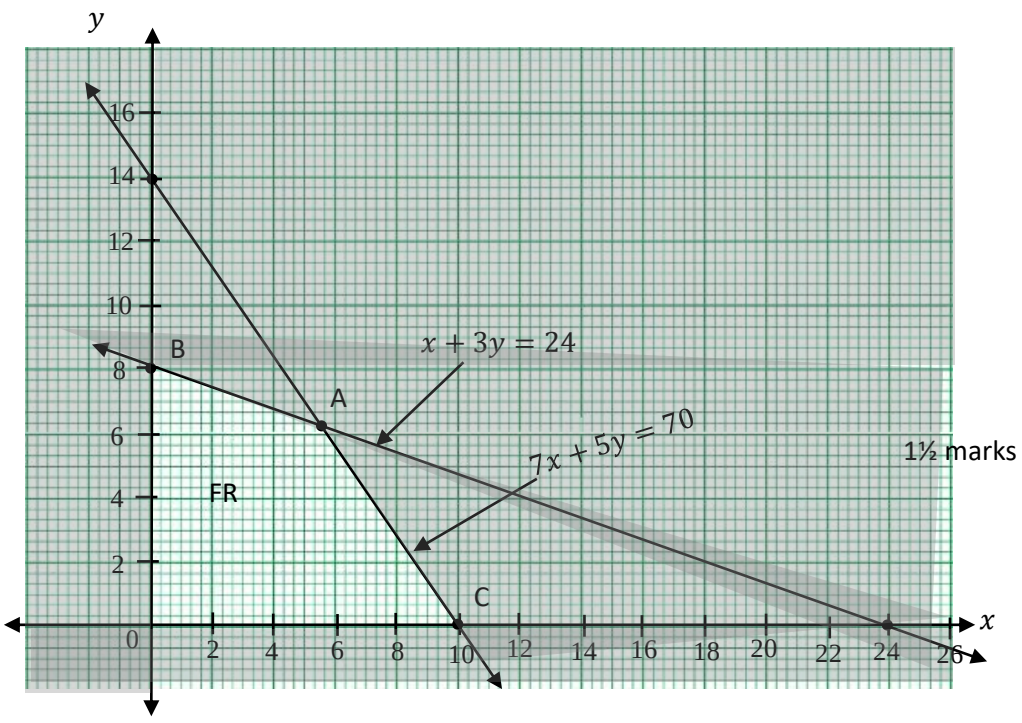
$$\begin{cases} 7x + 5y \leq 70 \\ x + 3y \leq 24 \\ x \geq 0, y \geq 0 \end{cases}$$ ……..$01\frac{1}{2}$ marks

Objective function $f(x, y) = 2\,000x + 1\,500y(maximize)$…….. $00\frac{1}{2}$ mark.

Constraint in intercept
$7x + 5y \leq 70$ *i.e* $(10, 0), (0, 14)$
$x + 3y \leq 24$ *i.e*$(24, 0), (0, 8)$

Graphing the constraint to obtain the decision-making region or feasible region

Figure 10



1½ marks

$$\begin{array}{l|l} 1 & 7x + 5y = 70 \\ 7 & x + 3y = 24 \end{array}$$

$$\begin{array}{r} 7x + 5y = 70 \\ 7x + 21y = 168 \end{array}$$

$$-\left|\begin{array}{l} 7x + 21y = 168 \\ 7x + 5y = 70 \end{array}\right.$$

$16y = 98$ $\frac{16y}{16} = \frac{98}{16}, y = 6\frac{1}{8}$

Substitute $6\frac{1}{8}$ $for\ y\ in\ x + 3y = 24$

$$x = 24 - 3y = 24 - 3 \times \frac{49}{8} = \frac{192-147}{8} = \frac{45}{8}$$

points	$f(x,y) = 2\,000x + 1\,500y$	Value in shillings
B(0,8)	$0\times 2000 + 8 \times 1500$	12 000
C (10 ,0)	$10 \times 2000 + 0 \times 1500$	20 000
A($6\frac{1}{8}, 5\frac{5}{8}$)i.e A(6,6)	$6 \times 2000 + 6 \times 1500$	21 000…….$1\frac{1}{2}$ marks

Therefore the industry should make 6 bars of soaps of grade A and 6 bars

soaps of grade B to earn the maximum profit of 21 000 shillings……$00\frac{1}{2}$ mark